全民英檢中級 GEPT Intermediate | 第 21 單元 | 科學與度量 · Science & Measurement

The Day Our Experiment Went Wrong

實驗出錯的那一天

Last Friday, our chemistry class moved from the classroom to the school laboratory for the first time this semester. Ms. Lin, who had worked as a chemist before becoming a teacher, told us that we would test how quickly different powders dissolve in water. Each group was given a small bottle of weak acid, three white compounds, and a worksheet with a formula for calculating the speed of the reaction.

My partner Kevin and I were confident at first. We had read the chapter twice, and we knew that an accurate result depended on measuring every gram carefully. However, Kevin poured too much powder into the beaker before I had finished weighing it. The mixture bubbled over the edge, and a sweet smell filled the room. Ms. Lin hurried over, but instead of being angry, she laughed and asked us what we thought had happened.

Through the lens of a magnifying glass, she showed us tiny bubbles that were still forming. The powder contained carbon, she explained, and when it met the acid, it released a gas that we could not see. Every bubble was made of atoms that had been rearranged. Our mistake had actually turned into a small discovery about how fast a reaction can happen when the amounts are wrong.

By the end of the lesson, we had redone the experiment three times and recorded the results with much greater accuracy. Walking home past the night market, I kept thinking about how electricity, gravity, and even the rice in my dinner are all made of the same tiny building blocks. Science, I realized, is not only something in a textbook; it is happening in every beaker, and in every kitchen, all the time.

重點單字 Key Vocabulary

單字	詞性	中文	例句
laboratory	*n.*	實驗室	The students put on goggles before entering the laboratory. 學生們進實驗室前先戴上護目鏡。

experiment	*n.*	實驗	The experiment proved that plants need light to grow. 這個實驗證明植物需要光才能生長。
chemist	*n.*	化學家	The chemist mixed two liquids and watched the color change. 那位化學家把兩種液體混合，觀察顏色的變化。
dissolve	*v.*	溶解	Sugar dissolves faster in hot tea than in cold tea. 糖在熱茶裡比在冷茶裡溶解得更快。
acid	*n.*	酸	Lemon juice contains a mild acid that can clean metal. 檸檬汁含有一種能清潔金屬的弱酸。
compound	*n.*	化合物	Water is a compound made of hydrogen and oxygen. 水是由氫和氧組成的化合物。
formula	*n.*	公式；配方	Use this formula to find the area of a circle. 用這個公式來求圓的面積。
accurate	*adj.*	準確的	The weather report was surprisingly accurate this week. 這週的天氣預報出乎意料地準確。
carbon	*n.*	碳	Diamonds and pencil lead are both made of carbon. 鑽石和鉛筆芯都是由碳構成的。
discovery	*n.*	發現	The discovery of penicillin saved millions of lives. 青黴素的發現拯救了數百萬人的生命。

片語 Phrases

at first	起初；一開始	At first I did not like the song, but now I love it. 起初我不喜歡這首歌，但現在我很愛。
depend on	取決於；依賴	Whether we go hiking depends on the weather. 我們去不去健行取決於天氣。
turn into	變成	The small argument turned into a big fight. 小小的爭執變成了一場大吵。

文法與句型 Grammar

S + had + p.p. ... before + S + V-ed

過去完成式（had + 過去分詞）表示在過去某個動作之前已經發生或完成的事。文中 Kevin poured too much powder before I had finished weighing it，強調「我還沒秤完」這件事在倒粉末之前仍在進行。

The movie had already started before we found our seats.

我們找到座位之前，電影就已經開始了。

instead of + V-ing / N

instead of 意思是「而不是…」，後面接名詞或動名詞，用來表示「沒有做 A，反而做了 B」。文中 instead of being angry, she laughed 表示老師沒有生氣，反而笑了。

Instead of taking the bus, we walked to the MRT station.

我們沒有搭公車，而是走路到捷運站。

讀完文章後，請到「皇家大衛英文閱讀集點站」完成本單元測驗，答對即可累積點數。